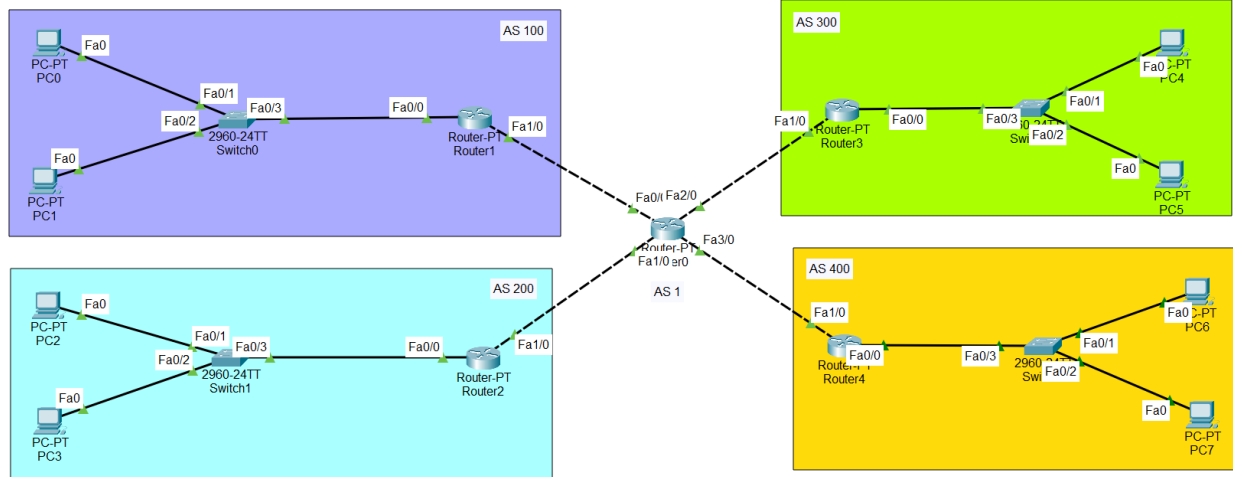


CNET Lab 15

Task 1:



```
Router(config-if) #
Router(config-if) #EXIT
Router(config) #router bgp 1
Router(config-router) #bgp router-id 5.5.5.5
Router(config-router) #neighbor 10.0.0.2 remote-as 100
Router(config-router) #neighbor 20.0.0.2 remote-as 200
Router(config-router) #neighbor 30.0.0.2 remote-as 300
Router(config-router) #neighbor 40.0.0.2 remote-as 400
Router(config-router) #network 10.0.0.0 mask 255.0.0.0
Router(config-router) #network 20.0.0.0 mask 255.0.0.0
Router(config-router) #network 30.0.0.0 mask 255.0.0.0
Router(config-router) #network 40.0.0.0 mask 255.0.0.0
Router(config-router) #exit
Router(config) #
```

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC3	Router2	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC0	Router1	ICMP		0.000	N	1	(edit)	(delete)
	Successful	Router3	Router0	ICMP		0.000	N	2	(edit)	(delete)
	Successful	Router1	Router4	ICMP		0.000	N	3	(edit)	(delete)

BGP Results:

```

Router#show ip protocols
Routing Protocol is "bgp 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  IGP synchronization is disabled
  Automatic route summarization is disabled
Neighbor(s):
  Address          FiltIn FiltOut DistIn DistOut Weight RouteMap
  10.0.0.2
  20.0.0.2
  30.0.0.2
  40.0.0.2
Maximum path: 1
  Routing Information Sources:
    Gateway          Distance      Last Update
    10.0.0.2          20            00:00:00
    20.0.0.2          20            00:00:00
    30.0.0.2          20            00:00:00
    40.0.0.2          20            00:00:00
  Distance: external 20 internal 200 local 200
Router#

```

```

Router#show ip route bgp
B   192.168.10.0/24 [20/0] via 10.0.0.2, 00:00:00
B   192.168.20.0/24 [20/0] via 20.0.0.2, 00:00:00
B   192.168.30.0/24 [20/0] via 30.0.0.2, 00:00:00
B   192.168.40.0/24 [20/0] via 40.0.0.2, 00:00:00

```

```

Router#show ip protocols
Routing Protocol is "bgp 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  IGP synchronization is disabled
  Automatic route summarization is disabled
Neighbor(s):
  Address          FiltIn FiltOut DistIn DistOut Weight RouteMap
  10.0.0.2
  20.0.0.2
  30.0.0.2
  40.0.0.2
Maximum path: 1
  Routing Information Sources:
    Gateway          Distance      Last Update
    10.0.0.2          20            00:00:00
    20.0.0.2          20            00:00:00
    30.0.0.2          20            00:00:00
    40.0.0.2          20            00:00:00
  Distance: external 20 internal 200 local 200
Router#show ip route bgp
B   192.168.10.0/24 [20/0] via 10.0.0.2, 00:00:00
B   192.168.20.0/24 [20/0] via 20.0.0.2, 00:00:00
B   192.168.30.0/24 [20/0] via 30.0.0.2, 00:00:00
B   192.168.40.0/24 [20/0] via 40.0.0.2, 00:00:00

```

```

- . . .

```

```

Router#show ip bgp neighbors
BGP neighbor is 10.0.0.2, remote AS 100, external link
  BGP version 4, remote router ID 1.1.1.1
  BGP state = Established, up for 00:10:42
  Last read 00:10:42, last write 00:10:42, hold time is 180, keepalive interval is 60 seconds
  Neighbor capabilities:
    Route refresh: advertised and received(new)
    Address family IPv4 Unicast: advertised and received
  Message statistics:
    InQ depth is 0
    OutQ depth is 0

      Sent      Rcvd
Opens:          1         1
Notifications:  0         0
Updates:        14         2
Keepalives:     11        11
Route Refresh:   0         0
Total:          26        14
Default minimum time between advertisements runs is 30 seconds

For address family: IPv4 Unicast
  BGP table version 13, neighbor version 6/0
--More--

```

```

Router#show ip bgp summary
BGP router identifier 5.5.5.5, local AS number 1
BGP table version is 13, main routing table version 6
12 network entries using 1584 bytes of memory
12 path entries using 624 bytes of memory
8/4 BGP path/bestpath attribute entries using 1104 bytes of memory
5 BGP AS-PATH entries using 120 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 3464 total bytes of memory
BGP activity 8/0 prefixes, 12/0 paths, scan interval 60 secs

Neighbor      V    AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
10.0.0.2      4   100     15      13     13    0    0 00:11:06      4
20.0.0.2      4   200     12       9     13    0    0 00:07:43      4
30.0.0.2      4   300     10       7     13    0    0 00:05:56      4
40.0.0.2      4   400     10       7     13    0    0 00:05:20      4

```

```

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BGP router identifier 5.5.5.5, local AS number 1
BGP table version is 13, main routing table version 6
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BGP using 3464 total bytes of memory
BGP activity 8/0 prefixes, 12/0 paths, scan interval 60 secs

Neighbor      V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
10.0.0.2       4   100     15      13       13   0    0 00:11:06      4
20.0.0.2       4   200     12       9       13   0    0 00:07:43      4
30.0.0.2       4   300     10       7       13   0    0 00:05:56      4
40.0.0.2       4   400      10       7       13   0    0 00:05:20      4

Router#show ip bgp
BGP table version is 13, local router ID is 5.5.5.5
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network        Next Hop           Metric LocPrf Weight Path
*> 10.0.0.0/8     0.0.0.0             0         0 32768 i
*                 10.0.0.2             0         0   100 i
*> 20.0.0.0/8     0.0.0.0             0         0 32768 i
*                 20.0.0.2             0         0   200 i
*> 30.0.0.0/8     0.0.0.0             0         0 32768 i
*                 30.0.0.2             0         0   300 i
*> 40.0.0.0/8     0.0.0.0             0         0 32768 i
*                 40.0.0.2             0         0   400 i
*> 192.168.10.0/24 10.0.0.2             0         0   100 i
*> 192.168.20.0/24 20.0.0.2             0         0   200 i
*> 192.168.30.0/24 30.0.0.2             0         0   300 i
*> 192.168.40.0/24 40.0.0.2             0         0   400 i

```